

APPLICATION OF FUZZY LOGIC IN FACE RECOGNITION



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FUZZY LOGIC

INTRODUCTION



Fuzzy Logic is an intelligent approach based on uncertainty and imprecise information. Unlike classical binary logic, which focuses strictly on true or false values, fuzzy logic allows intermediate values between 0 and 1. This flexibility makes it well suited for real-world systems and measurements, where precise data is often difficult to obtain.

In face recognition systems, uncertainty and variability are inherent due to factors such as lighting conditions, facial expressions, pose variations, and image quality. Fuzzy logic provides an effective framework for handling these ambiguities by allowing partial membership values rather than strict classifications. By modeling similarities between facial features in a gradual manner, fuzzy logic-based face recognition systems can achieve more robust and reliable performance in real-world scenarios.

FUZZY LOGIC

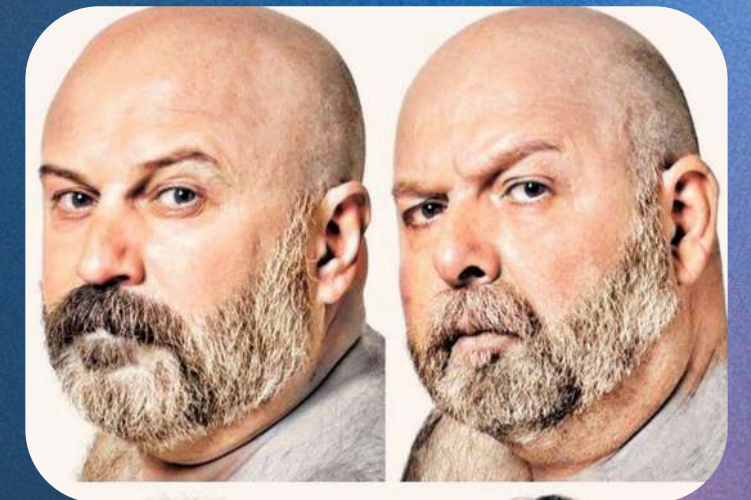
INTRODUCTION

 Fuzzy logic is widely used in systems that require human-like decision making in everyday life. In face recognition, humans do not identify faces using exact numerical values; instead, they rely on approximate judgments such as “very similar,” “slightly similar,” or “not similar.” Fuzzy logic models this reasoning by using linguistic variables and degrees of similarity rather than strict binary decisions.

slightly similar



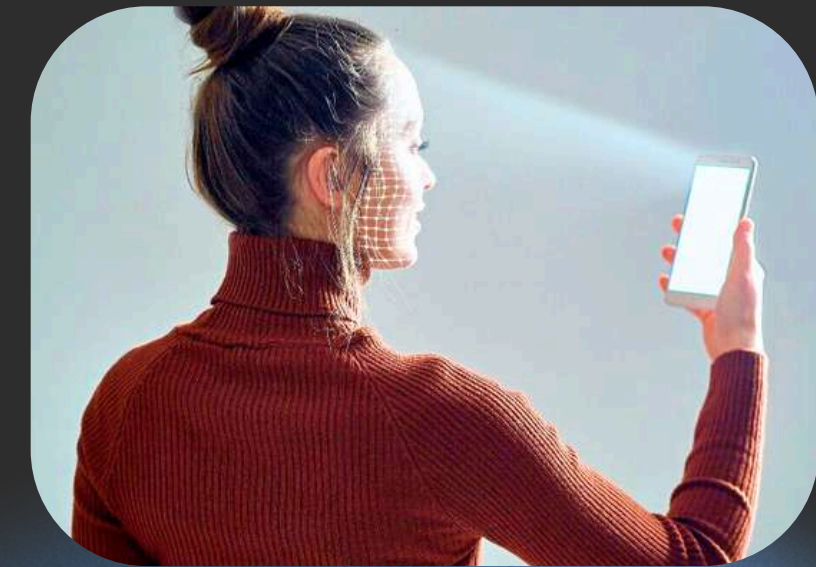
very similar



FUZZY LOGIC

INTRODUCTION

In real-world applications, face recognition systems must handle uncertainties caused by variations in lighting, facial expressions, pose, and image quality. Fuzzy logic has been effectively applied to address these challenges by allowing partial membership values in the decision-making process. This approach is commonly encountered in daily life, such as unlocking smartphones using facial recognition, automatically grouping people in photo album applications, identity verification in banking applications, automated systems used in driving license examinations, and the identification of suspects in criminal investigations. By incorporating fuzzy logic, face recognition systems achieve more flexible, robust, and reliable performance, particularly in environments where precise and consistent data cannot be guaranteed.



Feature	Classical Methods	Fuzzy Logic–Based Methods
Adaptability	Limited adaptability. Uses fixed thresholds and struggles with changes such as lighting, facial expression, or pose variations.	Highly adaptable. Easily handles changes in lighting, facial expressions, aging, and pose by modeling uncertainty like human perception.
Decision Making	Binary decision-based. A face is classified strictly as “match” or “no match.”	Gradual decision-making. Allows partial similarity levels such as “highly similar” or “moderately similar.”
Robustness	Sensitive to noise and low-quality images, leading to higher error rates.	Robust against noise, blur, and low-resolution images due to tolerance of imprecision.
Accuracy	Lower accuracy in real-world conditions where data is incomplete or inconsistent.	Higher accuracy in real-world scenarios by handling vague and imprecise information effectively.
Smoothness of Classification	Abrupt classification changes when thresholds are crossed.	Smooth and continuous classification without sudden decision jumps.
Real-World Performance	Performs well only under controlled conditions.	Performs reliably in uncontrolled, real-life environments such as surveillance and mobile authentication.

FIS METHODS

MAMDANI FIS

The Mamdani method uses fuzzy sets for both inputs and outputs and linguistic IF–THEN rules.

- Inference: min–max operators
- Defuzzification: Centroid method
- Advantage: High interpretability
- Disadvantage: Higher computational cost

SUGENO FIS

Sugeno FIS uses mathematical functions or constants as outputs.

- Output: Linear function or constant
- Defuzzification: Weighted average
- Advantage: Fast and efficient
- Disadvantage: Less intuitive

TSUKAMOTO FIS

Each rule in Tsukamoto FIS produces a crisp output using monotonic membership functions.

Final output: Weighted average

Advantage: Clear mathematical structure

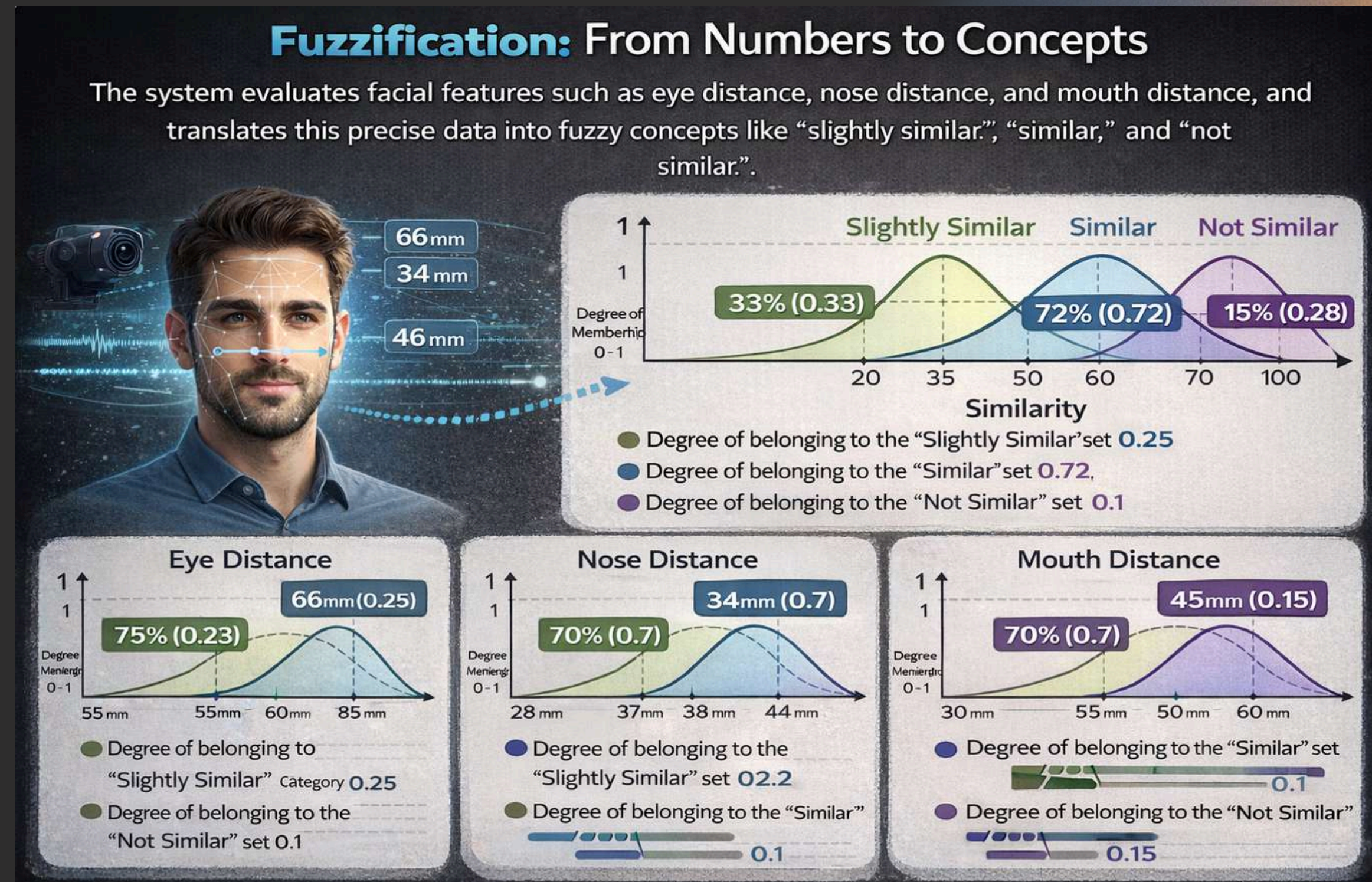
Disadvantage: Difficult rule design

LARSEN FIS

Larsen FIS is similar to Mamdani but uses multiplication instead of minimum for inference.

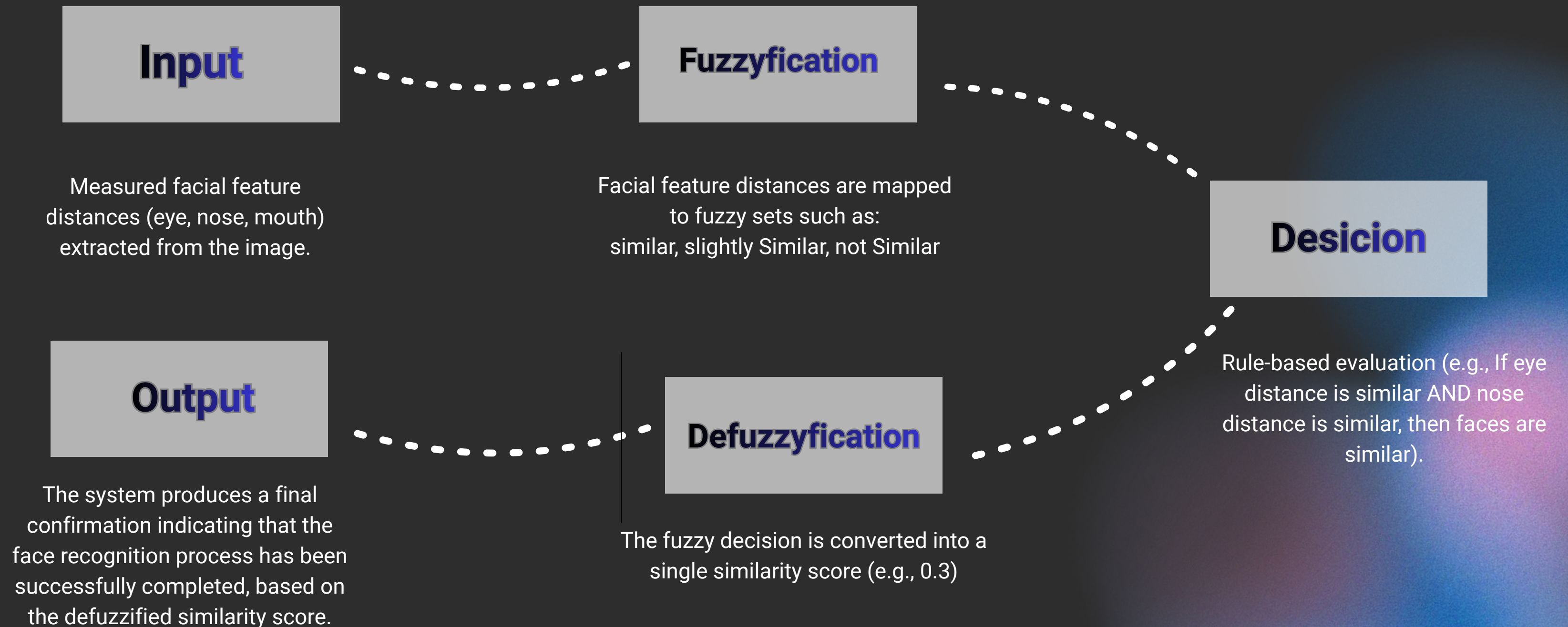
FUZZIFICATION

THE SYSTEM TAKES PRECISE DATA (SUCH AS EYE DISTANCE, NOSE DISTANCE, AND MOUTH DISTANCE) FROM FACIAL FEATURE MEASUREMENTS AND TRANSLATES THEM INTO HUMAN-LIKE CONCEPTS (FUZZY SETS) SUCH AS “SIMILAR,” “SLIGHTLY SIMILAR,” AND “NOT SIMILAR” TO DETERMINE FACIAL SIMILARITY.

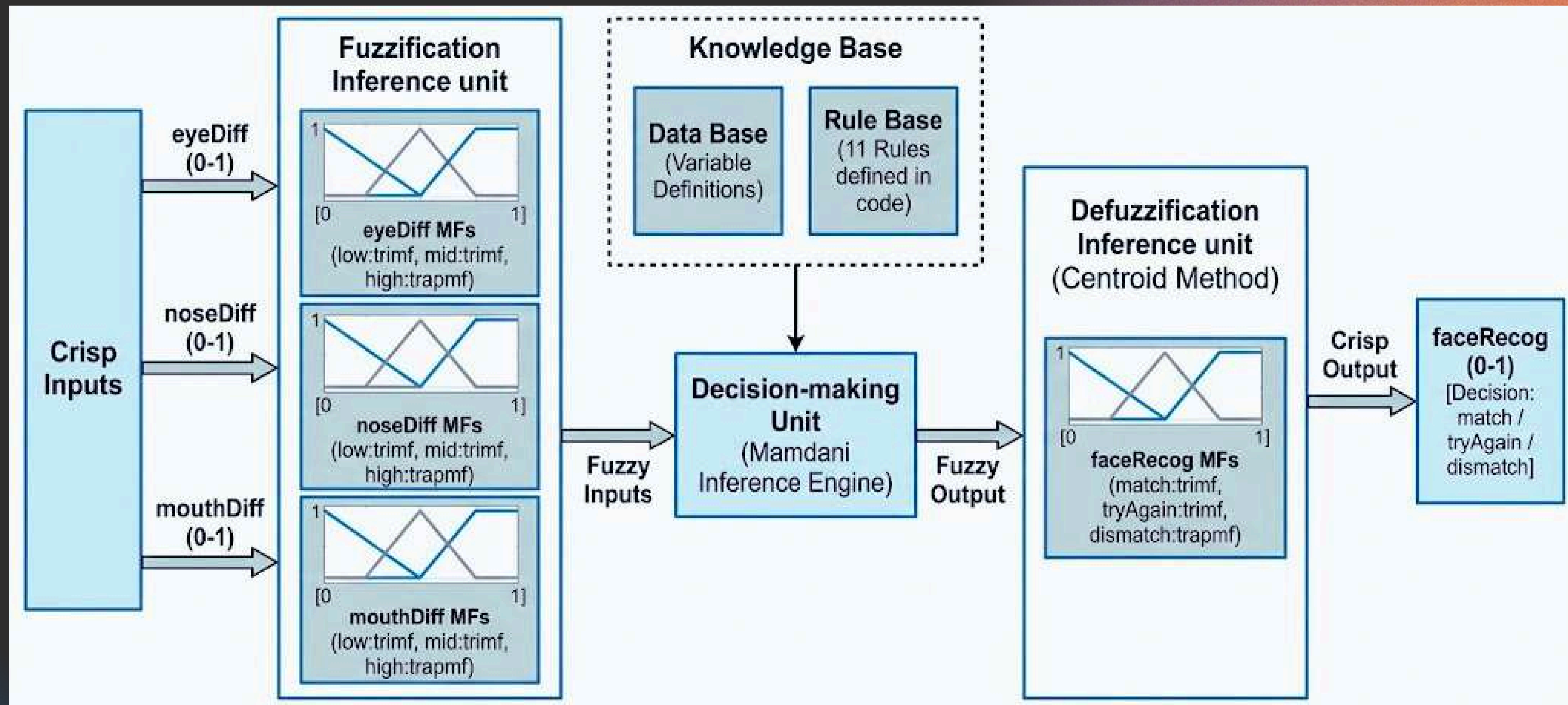


DEFUZZIFICATION

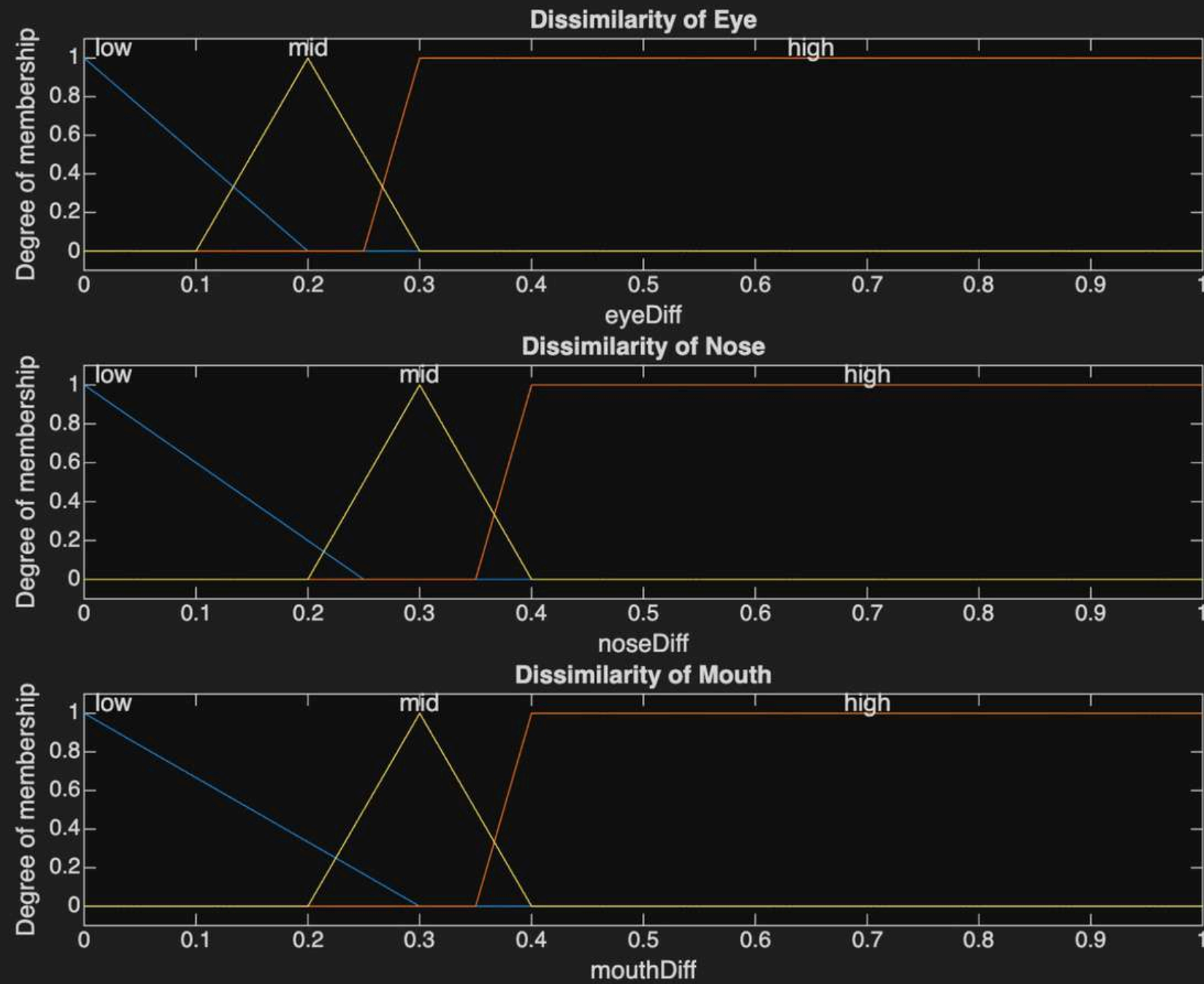
In the output phase, the system converts an abstract decision, such as “the faces are similar” or “slightly similar,” into a precise numerical similarity score that the facial recognition system can use for identification or verification.



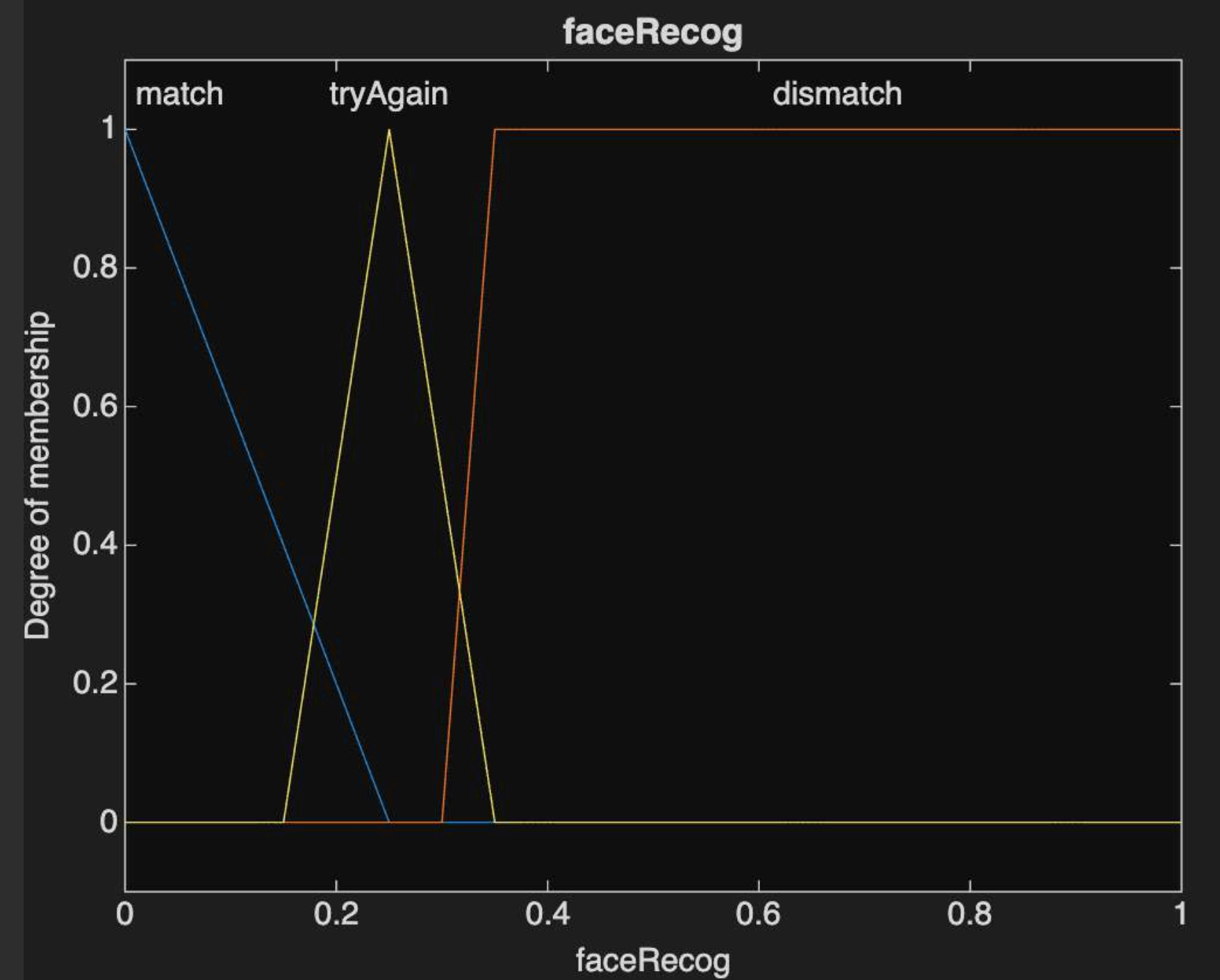
STRUCTURE OF THE FUZZY INFERENCE SYSTEM



INPUTS



OUTPUTS



Mapping Facial Dissimilarity to Matching Decisions

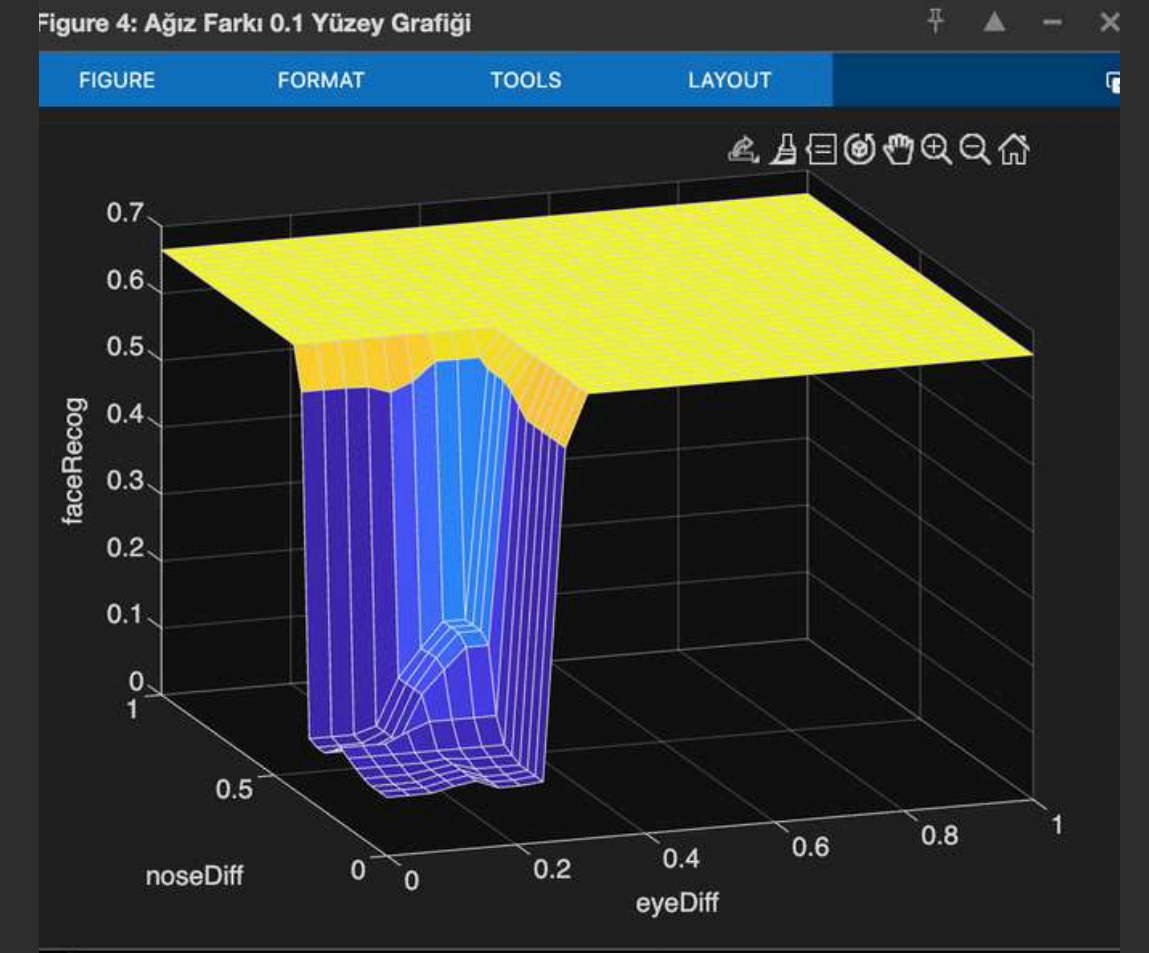
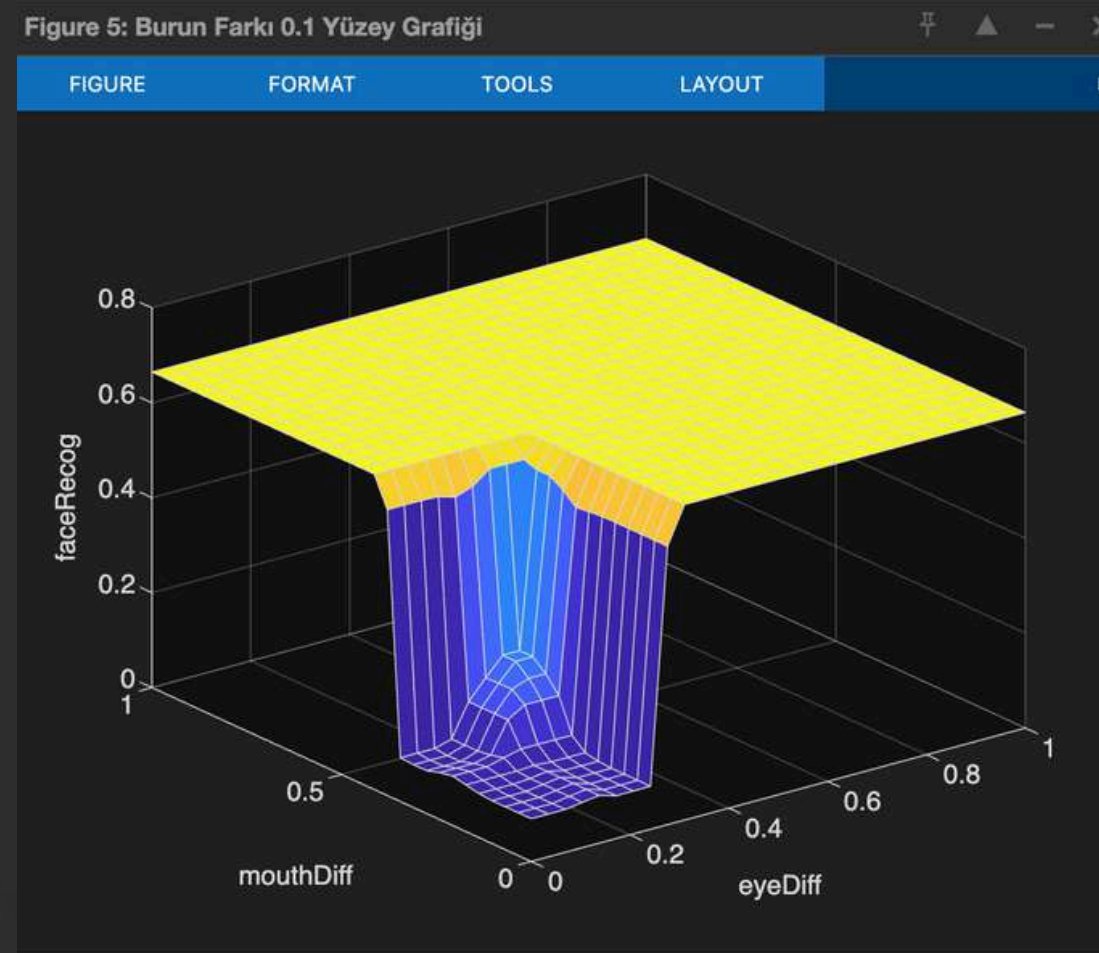
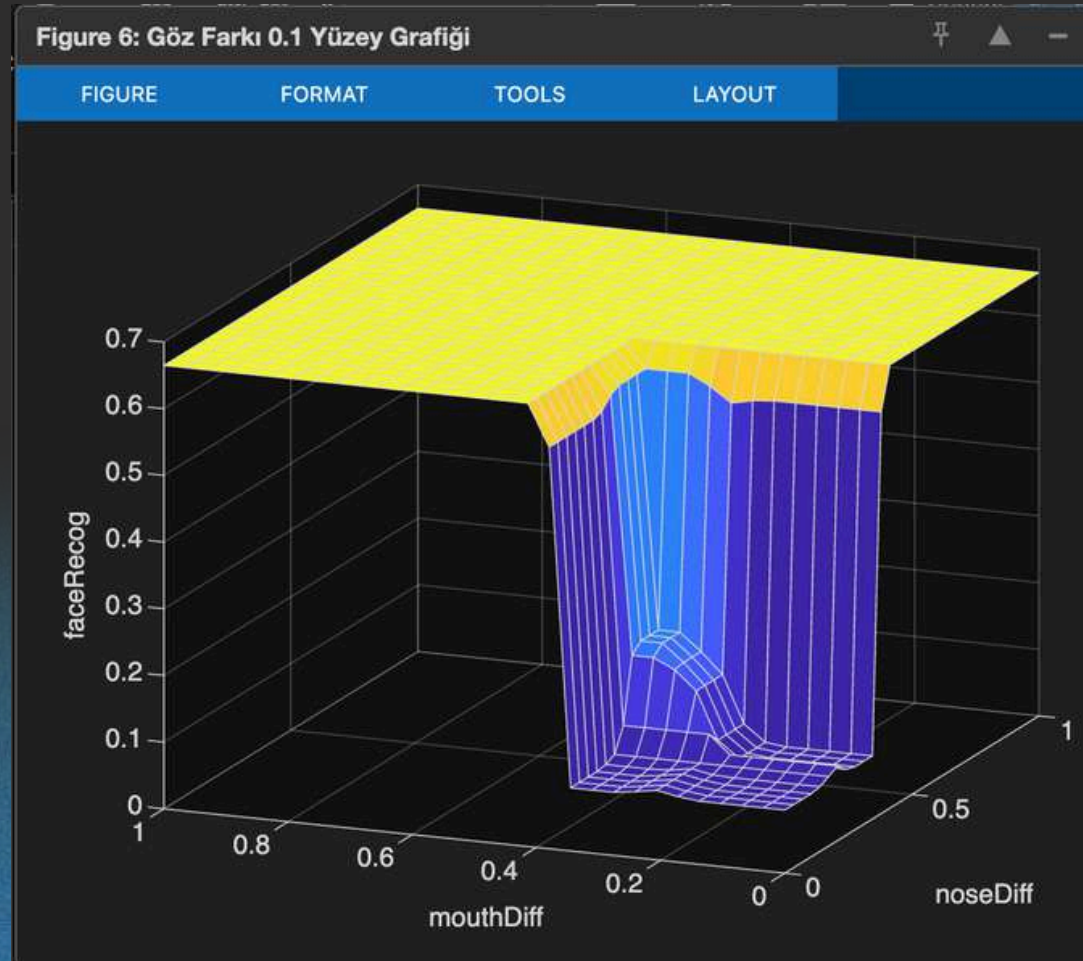
Dissimilarity Range	Decision Region	Explanation
0-0.2	Match	This range represents faces that are nearly identical. The facial features show minimal dissimilarity, indicating a very high likelihood that both images belong to the same person. The interval is kept narrow to reduce false matches.
0.2-0.4	Try Again	This region corresponds to uncertain cases where facial similarity is ambiguous. The system cannot confidently decide whether the faces belong to the same person, therefore a re-evaluation or additional verification
>0.4	Dismatch	In this range, the facial feature differences are significantly high, which statistically indicates that the images belong to different individuals. Hence, the decision is directly classified as a mismatch.

RULES

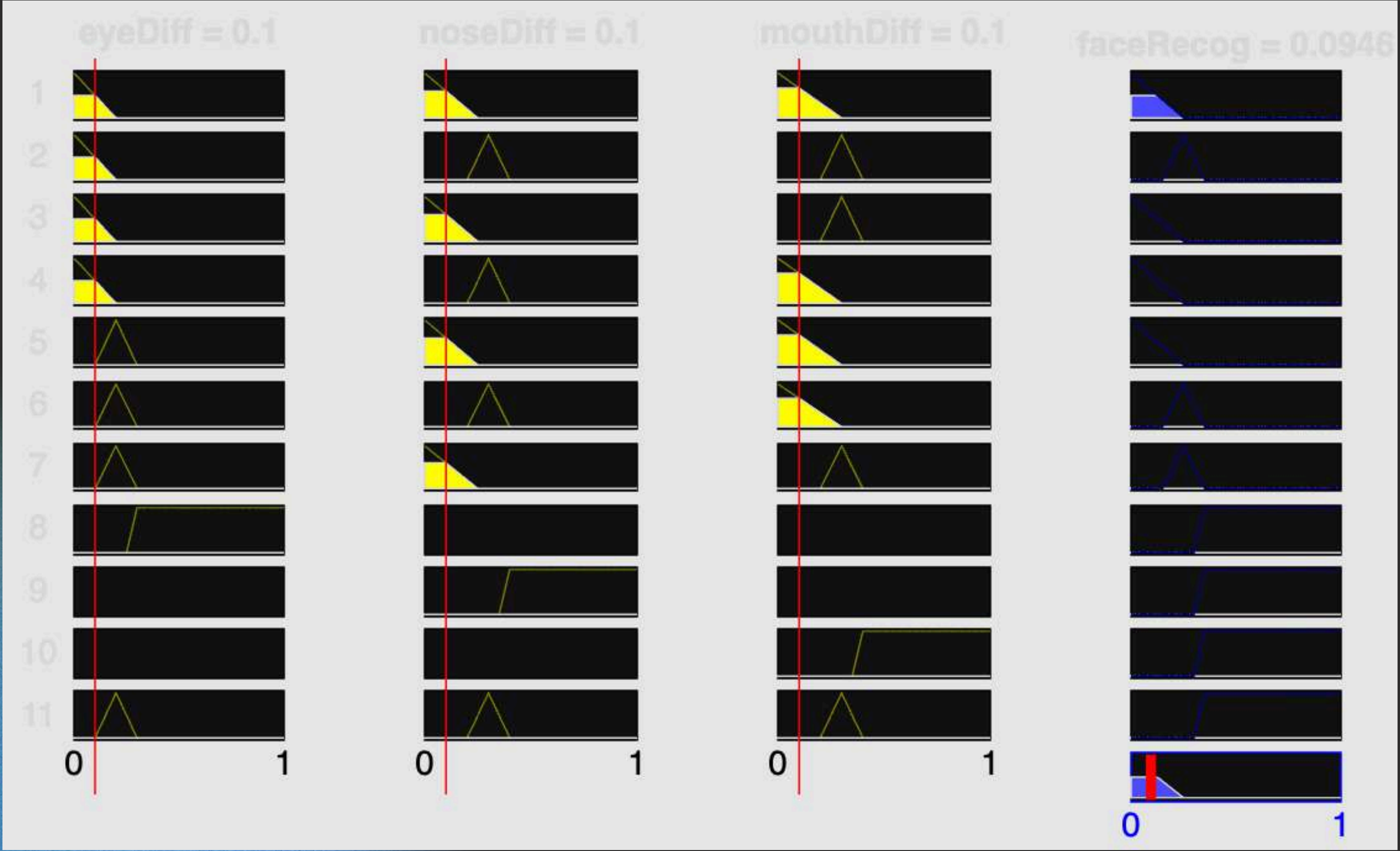
	Rule
1	If (eyeDiff is low) and (noseDiff is low) and (mouthDiff is low) then (faceRecog is match) (1)
2	If (eyeDiff is low) and (noseDiff is mid) and (mouthDiff is mid) then (faceRecog is tryAgain) (1)
3	If (eyeDiff is low) and (noseDiff is low) and (mouthDiff is mid) then (faceRecog is match) (1)
4	If (eyeDiff is low) and (noseDiff is mid) and (mouthDiff is low) then (faceRecog is match) (1)
5	If (eyeDiff is mid) and (noseDiff is low) and (mouthDiff is low) then (faceRecog is match) (1)
6	If (eyeDiff is mid) and (noseDiff is mid) and (mouthDiff is low) then (faceRecog is tryAgain) (1)
7	If (eyeDiff is mid) and (noseDiff is low) and (mouthDiff is mid) then (faceRecog is tryAgain) (1)
8	If (eyeDiff is high) then (faceRecog is mismatch) (1)
9	If (noseDiff is high) then (faceRecog is mismatch) (1)
10	If (mouthDiff is high) then (faceRecog is mismatch) (1)
11	If (eyeDiff is mid) and (noseDiff is mid) and (mouthDiff is mid) then (faceRecog is mismatch) (1)

CASE STUDY 1

FOR $\text{eyeDiff}=0.1, \text{noseDiff}=0.1, \text{mouthDiff}=0.1$ respectively:



FUZZY INFERENCE RESULTS FOR FIXED INPUT VALUE (0.1)



CASE STUDY 2

FOR $\text{eyeDiff} = 0.3, \text{noseDiff} = 0.3, \text{mouthDiff} = 0.3$ respectively:

Figure 5: Surface Graph for $\text{eyeDiff}=0.3$

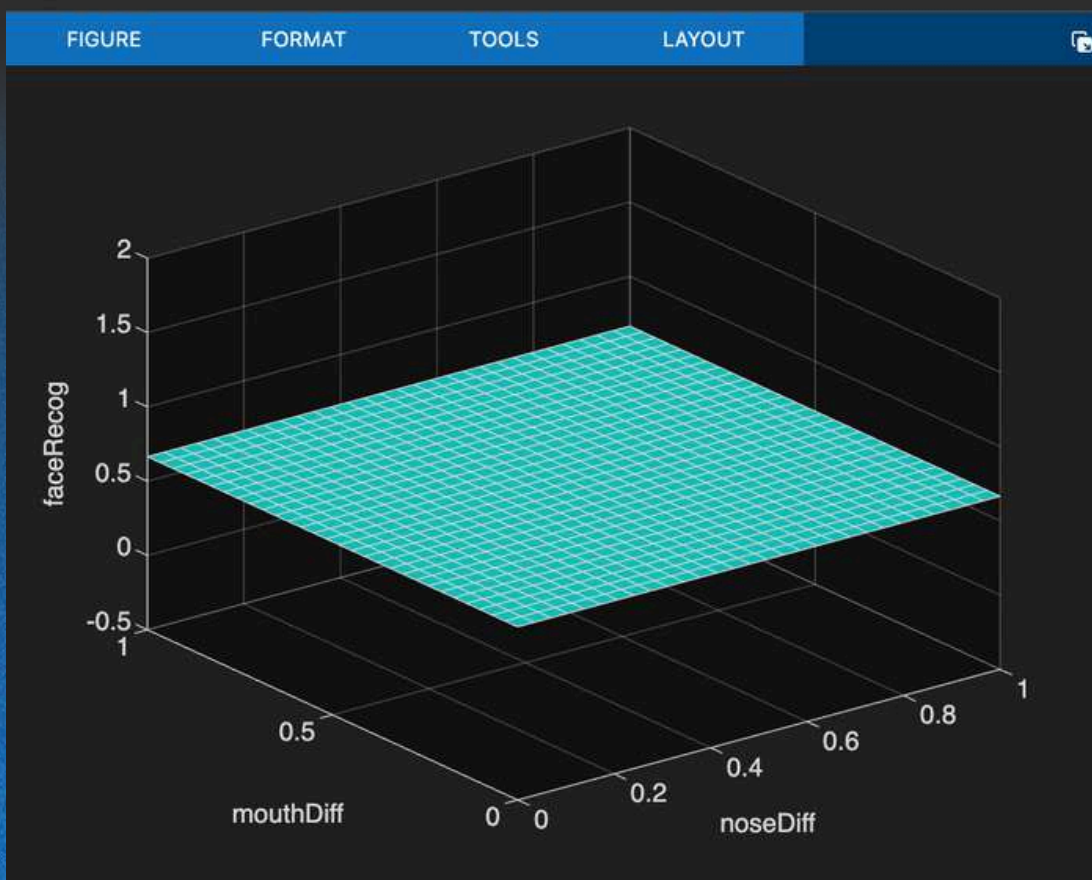


Figure 4: Surface Graph for $\text{noseDiff}=0.3$

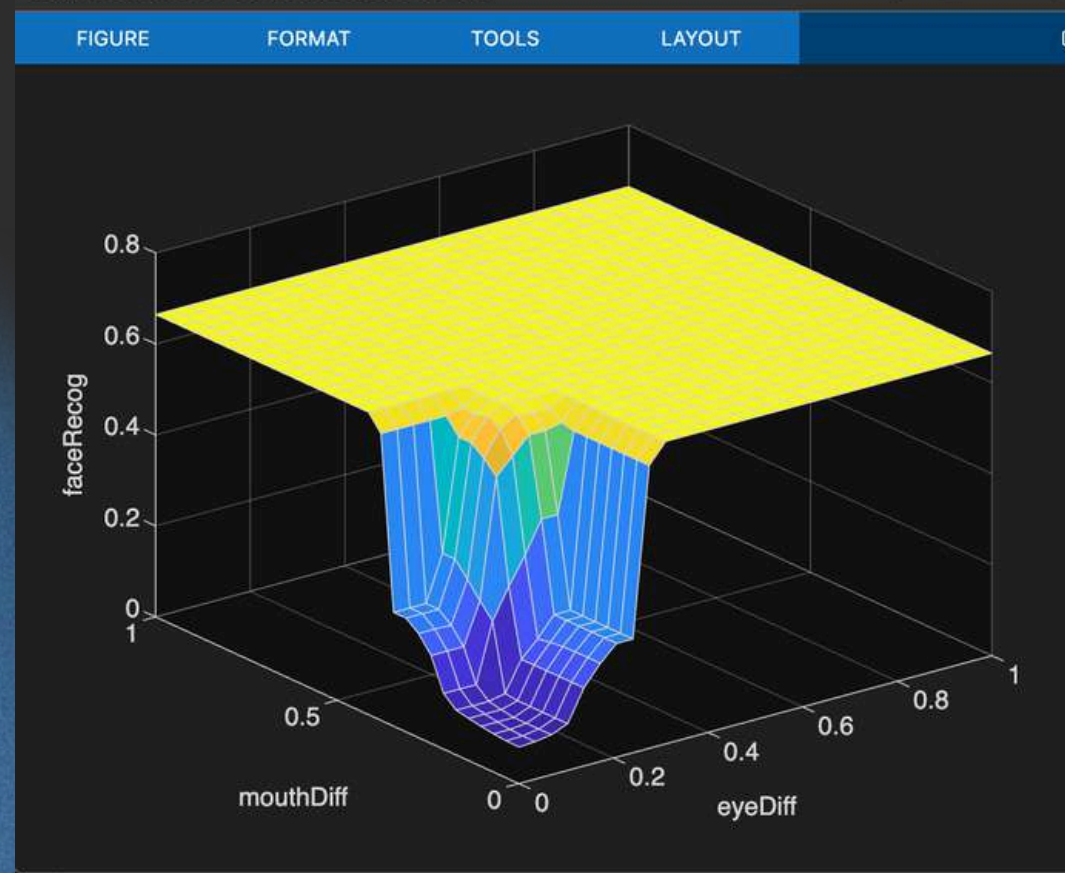
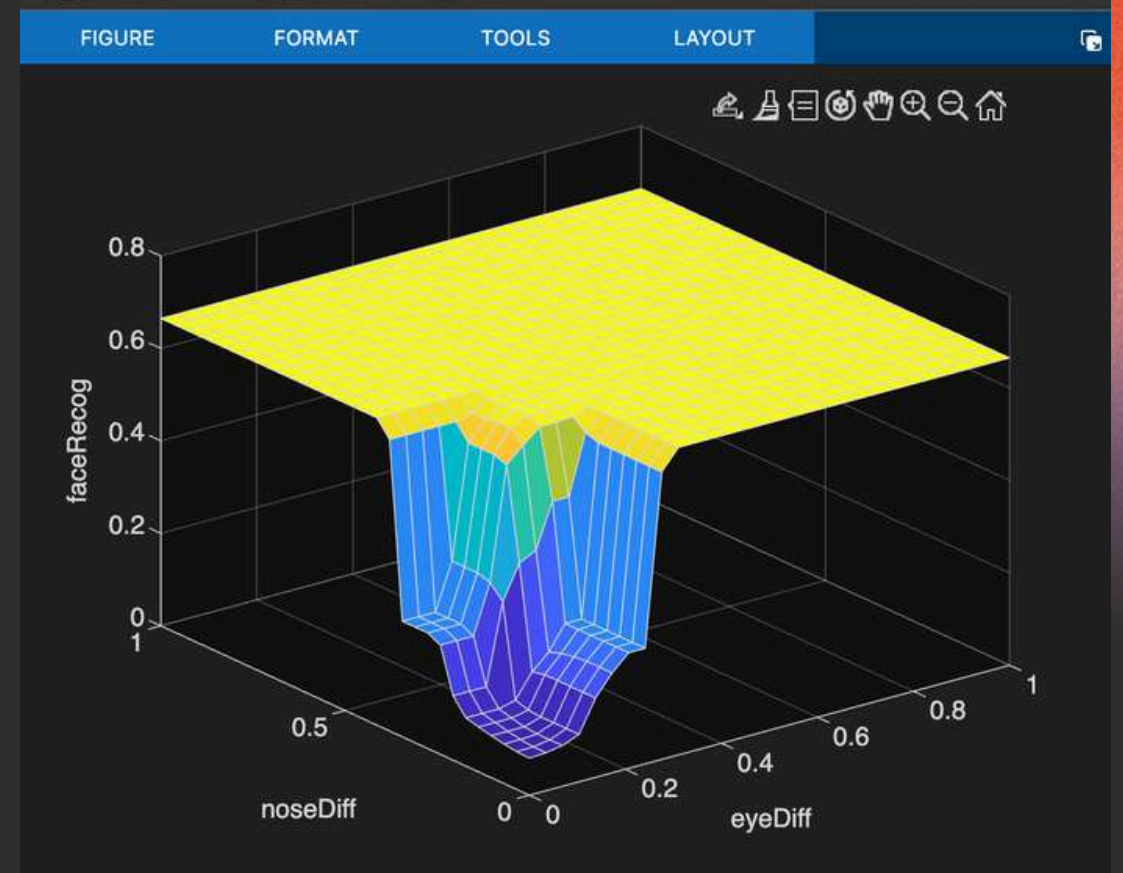
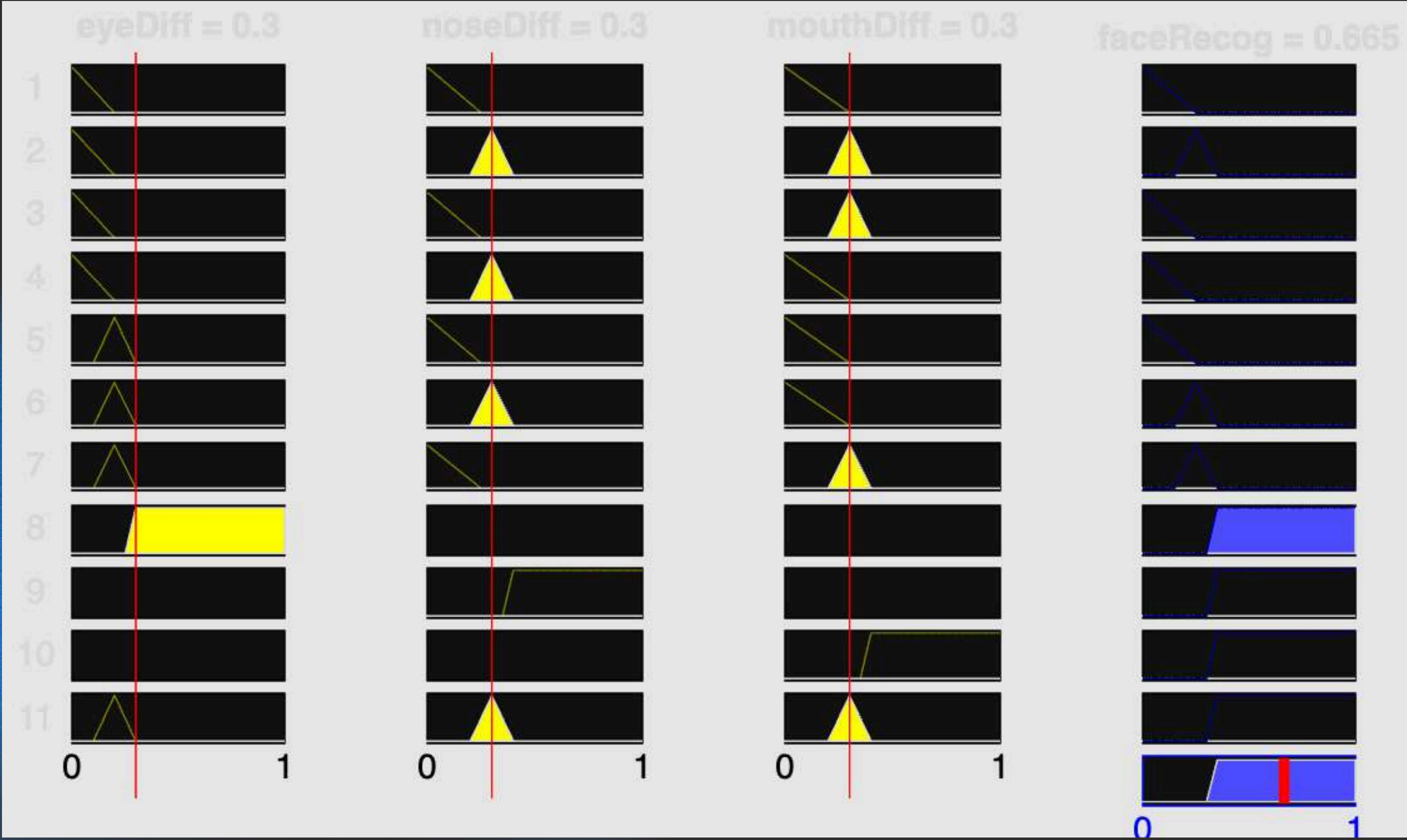


Figure 3: Surface Graph for $\text{mouthDiff}=0.3$

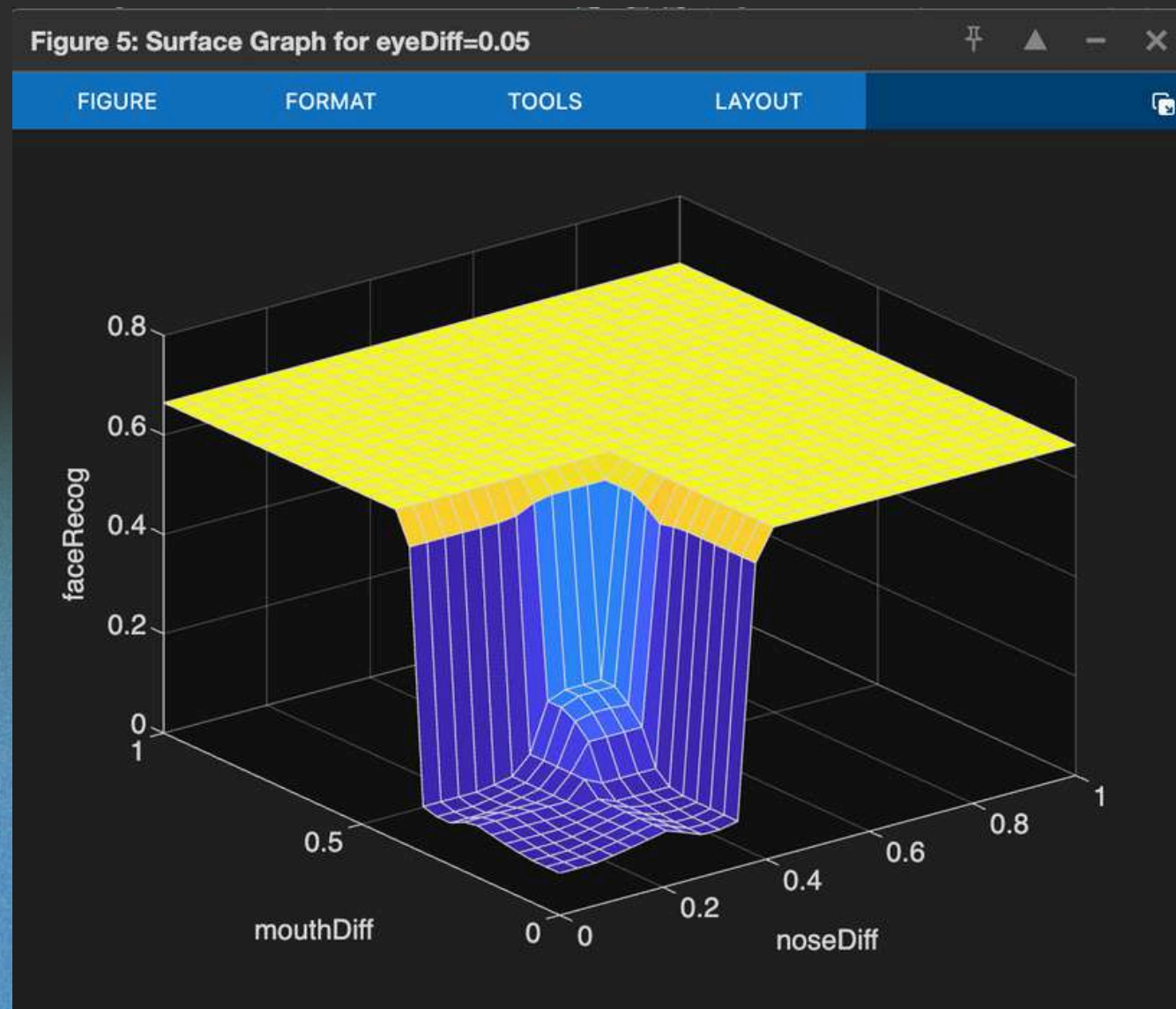


FUZZY INFERENCE RESULTS FOR FIXED INPUT VALUE (0.3)

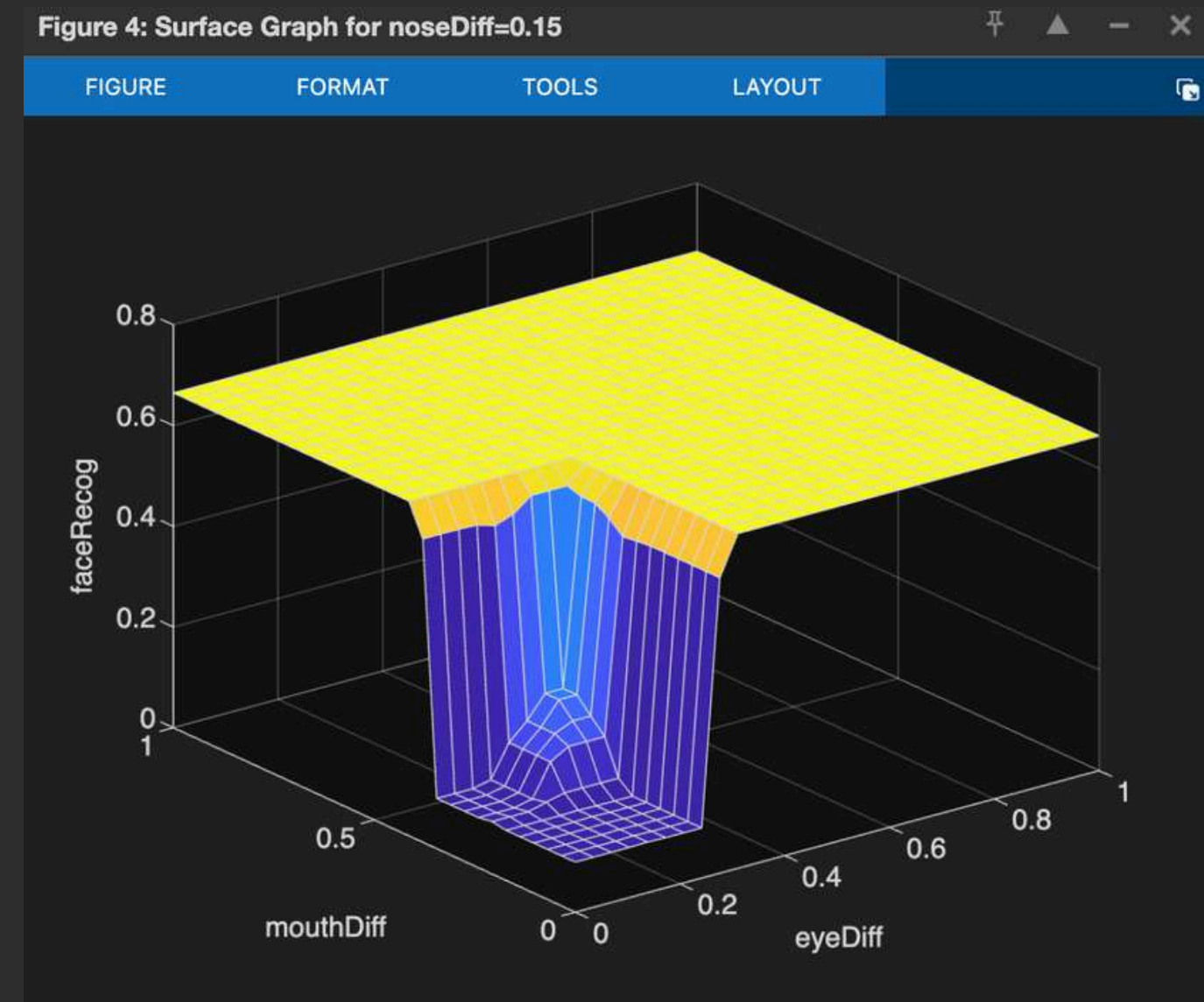


EXAMPLES

1 FOR $\text{eyeDiff} = 0.05$

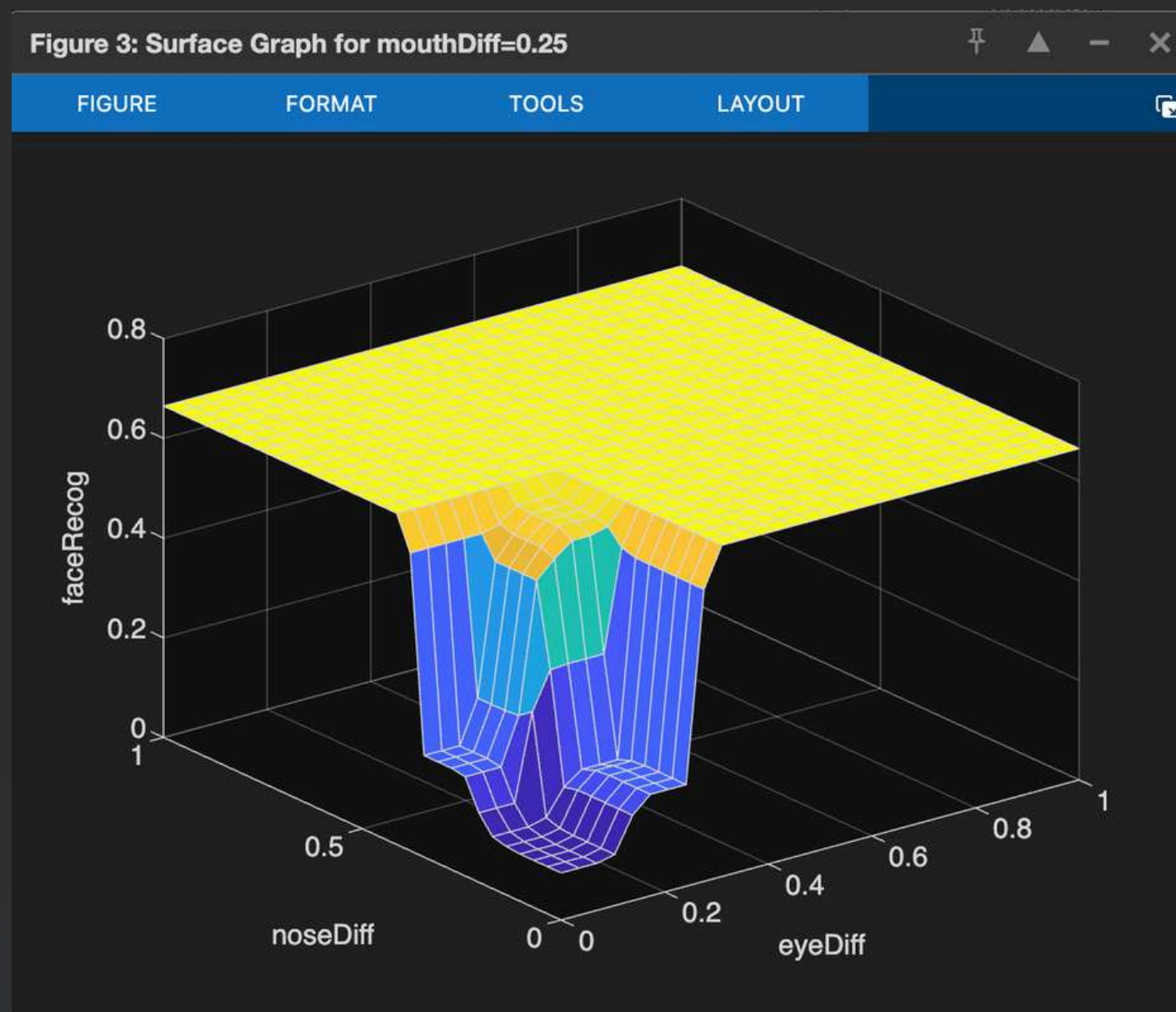


2 FOR $\text{noseDiff} = 0.15$

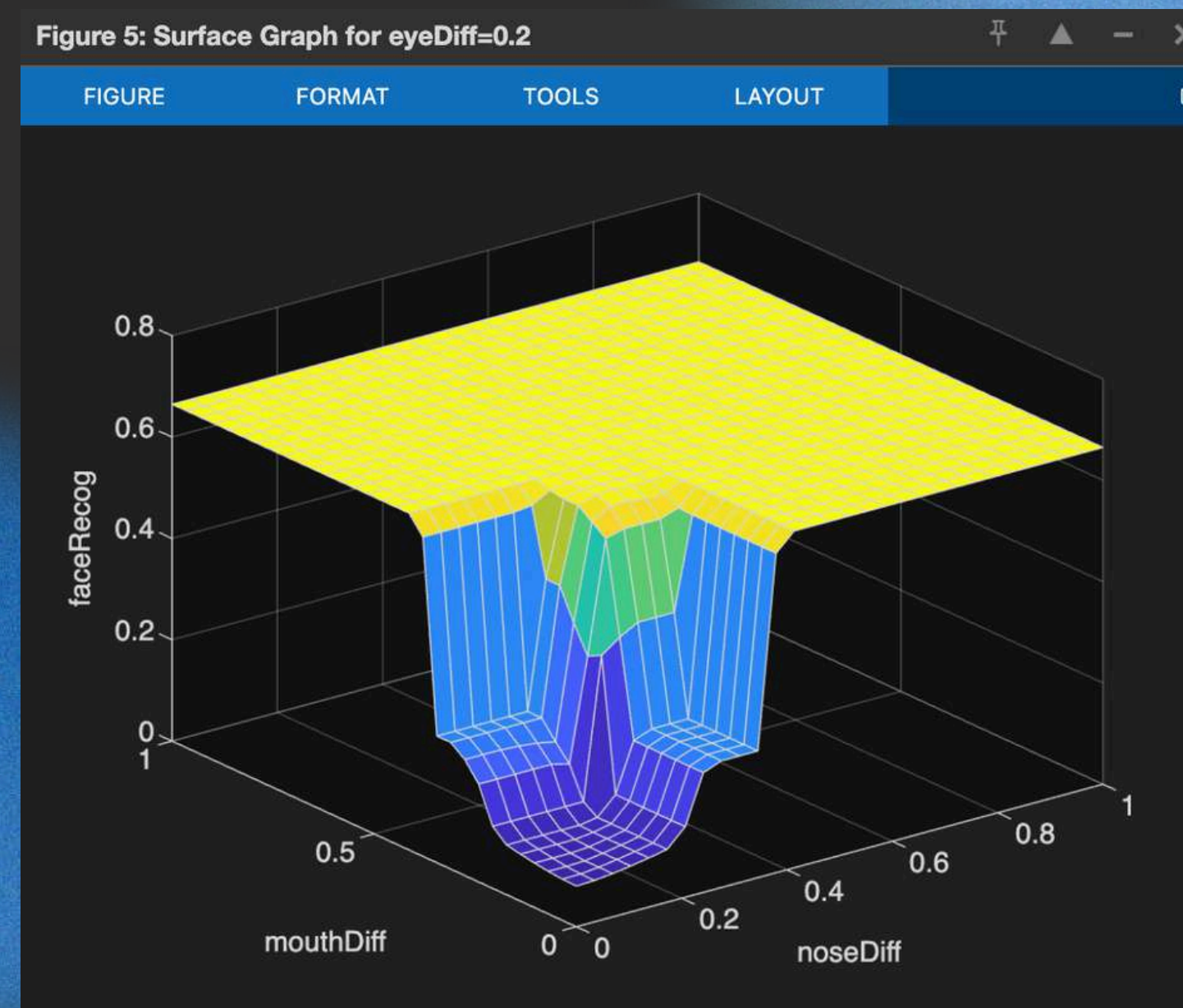


3

FOR mouthDiff = 0.25

**4**

FOR eyeDiff = 0.2



BEST MATCHING OUTPUT BASED ON FUZZY RULES



THANK YOU

FOR LISTENING